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Is metformin a wonder drug?

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Ever wonder why certain medications are called "wonder drugs"?

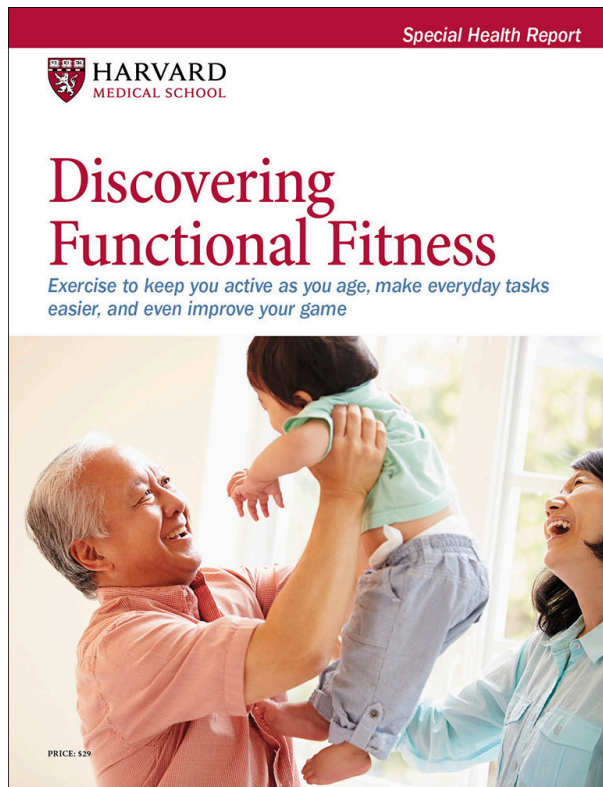
Sometimes it's because of the tremendous health benefits the drug provides for a particular condition, like insulin for type 1 diabetes or antibiotics for pneumonia. Or, it might be because the drug is good for many different conditions: aspirin has often been called a wonder drug because it can relieve pain, treat or prevent cardiovascular disease, and even prevent cancer.

Could metformin be joining this list? It's approved in the US to treat type 2 diabetes when used with diet and exercise by people ages 10 and older. But in recent years, interest has grown regarding its potential to prevent or treat a variety of other conditions, including aging. Yes, aging. If that's true, "wonder drug" might be an understatement.

What is metformin?

The history of metformin goes back hundreds of years. In Europe, the medicinal herb Galega officinalis was popular for digestive health and to treat urinary problems and other ailments. In 1918 a scientist discovered that one of its ingredients, guanidine, could lower blood sugar. Medicines containing guanidine, such as metformin and phenformin, were developed to treat diabetes. But they fell out of favor due to serious side effects caused by phenformin, and by the discovery of insulin.

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Metformin was rediscovered decades later and approved as a treatment for diabetes in Europe in the 1950s. It wasn't until 1995 that the FDA approved it for use in the US. It has since become the most widely prescribed medication for people with diabetes who cannot control their blood sugar through diet and exercise alone.

Metformin's benefits may extend far beyond diabetes

For decades we've known that metformin does more than just help lower blood sugar in people with diabetes. It also offers them cardiovascular benefits, including lower rates of death due to cardiovascular disease. And it sometimes helps people with diabetes lose excess weight.

Metformin may also have health benefits for people who don't have diabetes. Doctors have long prescribed it off-label - that is, to treat conditions outside its approved use, including:

- **Prediabetes.** People with prediabetes have elevated blood sugar that isn't yet high enough to qualify as diabetes. Metformin may delay the onset of diabetes or even prevent it among people with prediabetes.
- **Gestational diabetes.** Pregnant women may develop elevated blood sugar that returns to normal after delivery. Metformin can help control blood sugar during pregnancy in such women.
- **Polycystic ovary syndrome (PCOS).** This disorder tends to affect young women whose ovaries develop multiple cysts. Menstrual irregularities and fertility problems are common. Although the results of clinical studies are mixed, metformin has been prescribed for years for women with PCOS to help with menstrual regulation, fertility, and elevated blood sugar.
- **Weight gain from antipsychotic medicines.** Antipsychotics are powerful medications prescribed for psychiatric diseases such as schizophrenia. One common side effect is significant weight gain. Metformin may lessen weight gain among some people taking these drugs.

In addition, researchers are investigating the potential of metformin to

- Lower the risk of cancer in persons with type 2 diabetes. These include cancers of the breast, colon, and prostate.
- **Lower risks for dementia and stroke.** Some studies have noted less cognitive decline and a lower rate of dementia, as well as a lower rate of stroke, among people with diabetes taking metformin compared

with those who were not taking it.

- **Slow aging, prevent age-related disease, and increase lifespan.** Preliminary studies suggest that metformin may actually slow aging and increase life expectancy, possibly by improving the body's responsiveness to insulin, antioxidant effects, and improving blood vessel health.

Because the vast majority of research regarding metformin included only people with diabetes or prediabetes, it's unclear whether these potential benefits are limited to people with those conditions, or whether people *without* diabetes may derive benefit as well.

What about side effects?

The safety profile for metformin is quite good. Side effects include nausea, stomach upset, or diarrhea; these tend to be mild. More serious side effects are rare. They include severe allergic reactions and a condition called lactic acidosis, a buildup of lactic acid in the bloodstream. The risk for this is higher among people with significant kidney disease, so doctors tend to avoid prescribing metformin for them.

The bottom line

Metformin is a first-line treatment for type 2 diabetes, according to current diabetes guidelines. It's relatively inexpensive and its potential side effects are well understood.

If you have diabetes and need metformin to help lower your blood sugar, its other potential health benefits are a wonderful - not harmful - side effect. And if you don't have diabetes? Well, its role in preventing or treating diseases, and possibly even slowing aging and extending life expectancy, is much less clear.

While the research so far is promising, we need more compelling evidence before endorsing its widespread use for people without diabetes. But, for clinical researchers hoping to repurpose an old medicine as a new wonder drug, metformin would seem like a great place to start.

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Dr. Robert H. Shmerling is the former clinical chief of the division of rheumatology at Beth Israel Deaconess Medical Center (BIDMC), and is a current member of the corresponding faculty in medicine at Harvard Medical School. ... [See Full Bio](#)

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